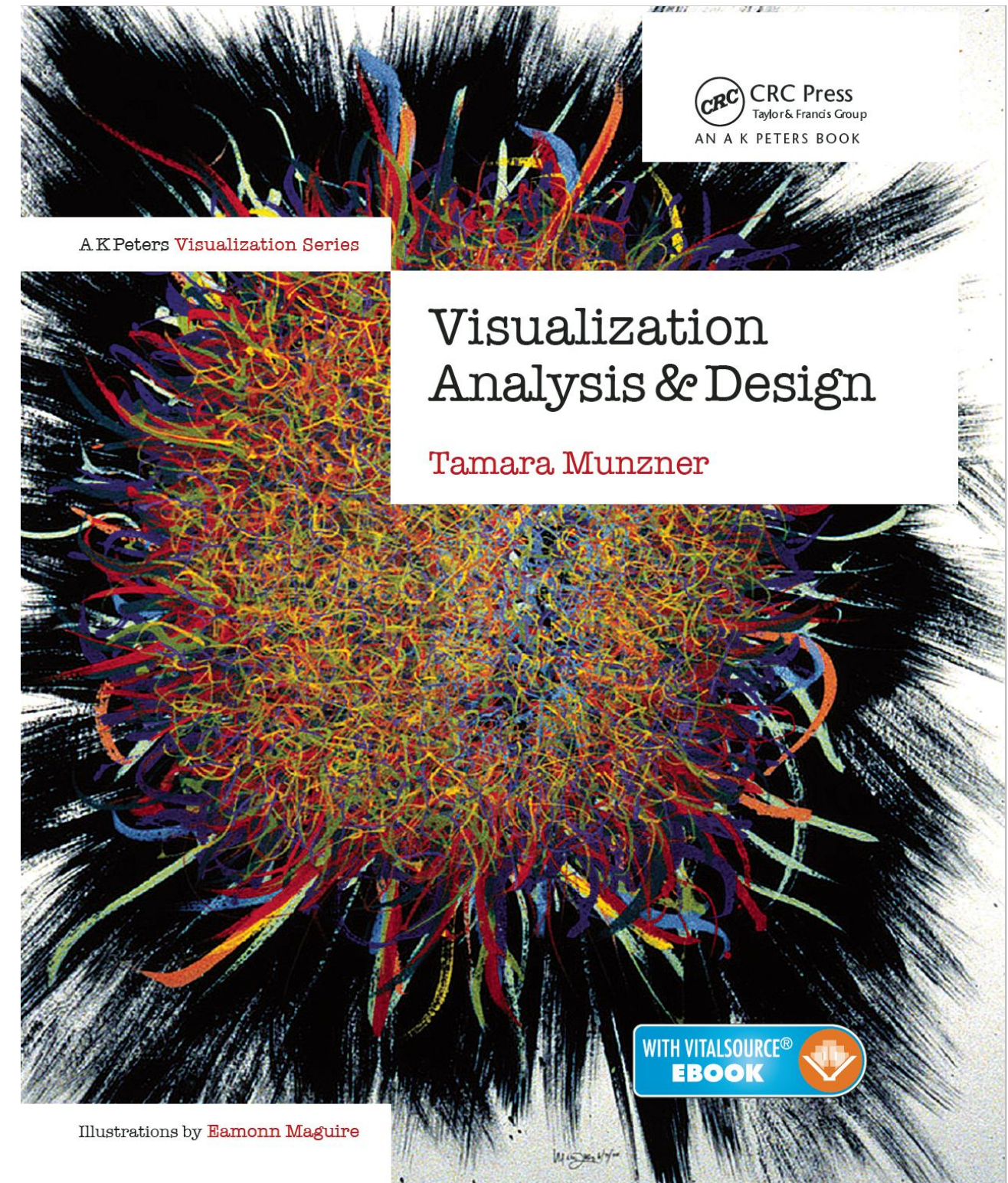


# CDS505

## Data Visualization & Visual Analytics

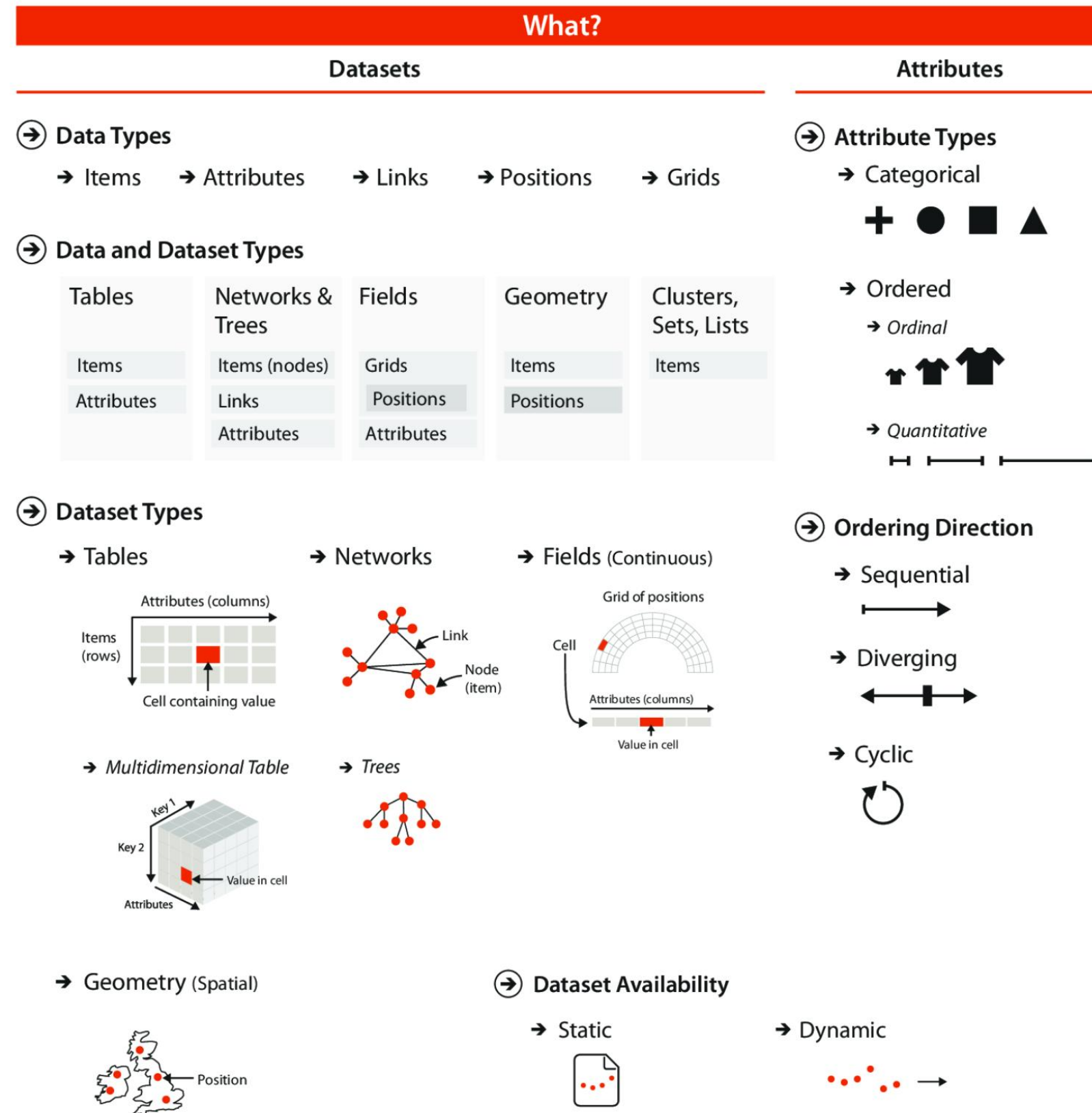
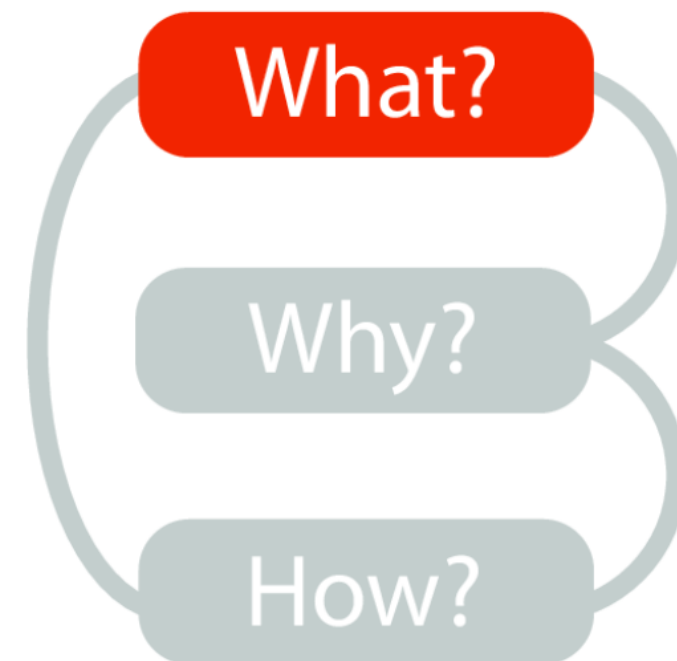
### Tables – *Visualizing Tabular Data*

*Nur Intan Raihana Ruhaiyem, PhD.  
School of Computer Sciences  
Universiti Sains Malaysia*

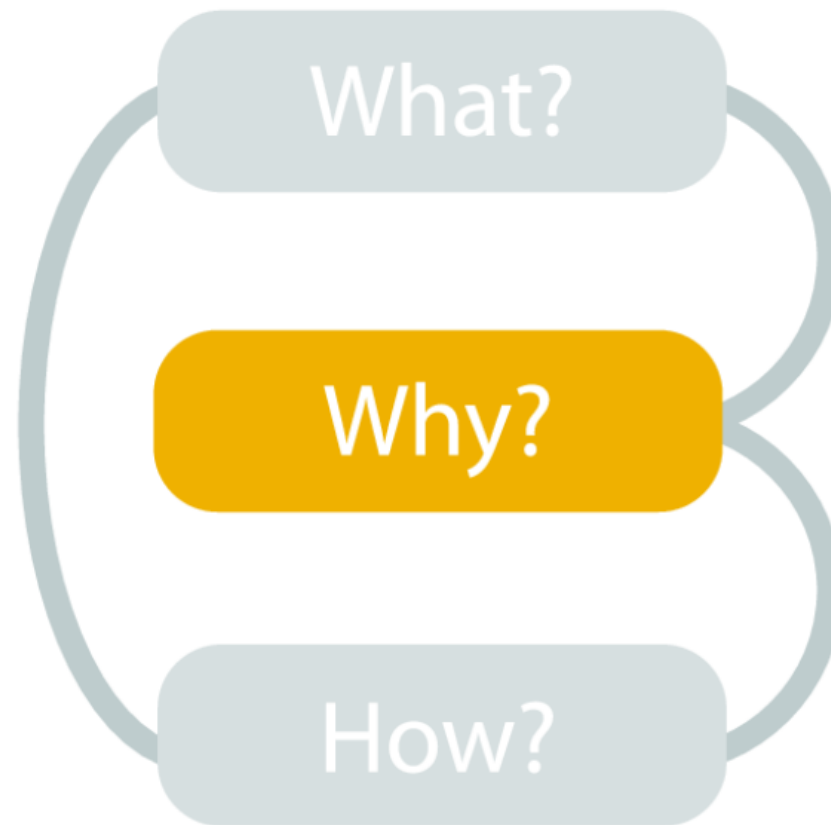


- ✓ Tables - Introduction
- ✓ Keys and Values – Scatterplots, Bar Charts, Heatmaps
- ✓ Quantitative Values
- ✓ Categorical Regions
- ✓ Spatial Axis Orientation: Rectilinear & Parallel Layouts
- ✓ Spatial Axis Orientation: Radial Layouts
- ✓ Spatial Layout Density

# Recap What-Why-How



# Recap What-Why-How



## Why?

### 👉 Actions

#### ➔ Analyze

##### ➔ Consume

##### ➔ Discover



##### ➔ Present



##### ➔ Enjoy



##### ➔ Produce

##### ➔ Annotate



##### ➔ Record



##### ➔ Derive



#### ➔ Search

|                  | Target known | Target unknown |
|------------------|--------------|----------------|
| Location known   | ••• Lookup   | ••• Browse     |
| Location unknown | <•••> Locate | <•••> Explore  |

#### ➔ Query

##### ➔ Identify



##### ➔ Compare



##### ➔ Summarize



### 🎯 Targets

#### ➔ All Data

##### ➔ Trends



##### ➔ Outliers



##### ➔ Features



#### ➔ Attributes

##### ➔ One

##### ➔ Distribution



##### ➔ Extremes

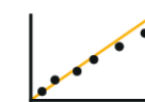


##### ➔ Many

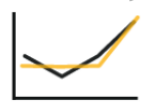
##### ➔ Dependency



##### ➔ Correlation

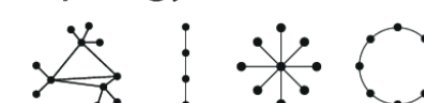


##### ➔ Similarity

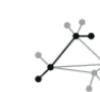


#### ➔ Network Data

##### ➔ Topology



##### ➔ Paths



#### ➔ Spatial Data

##### ➔ Shape



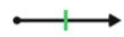
# Recap What-Why-How

## How?

### Encode

#### ➔ Arrange

➔ Express



➔ Separate



➔ Order



➔ Align



➔ Use



#### ➔ Map

from **categorical** and **ordered** attributes

➔ Color

➔ Hue



➔ Saturation



➔ Luminance



➔ Size, Angle, Curvature, ...



➔ Shape



➔ Motion

Direction, Rate, Frequency, ...



### Manipulate

#### ➔ Change



#### ➔ Select

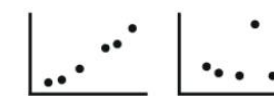


#### ➔ Navigate

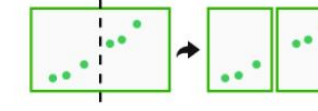


### Facet

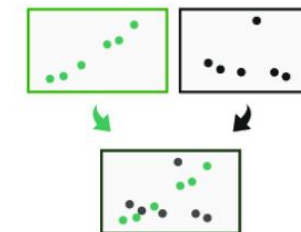
#### ➔ Juxtapose



#### ➔ Partition



#### ➔ Superimpose



### Reduce

#### ➔ Filter



#### ➔ Aggregate



#### ➔ Embed



What?

Why?

How?

## WHY ARRANGE?

Space is the most important channel.

## HOW TO ARRANGE?

- Quantitative data
  - Express
- Categorical data
  - Separate, order, align

### Encode

#### ➔ Arrange

➔ Express



➔ Order



➔ Separate



➔ Align



## Visual encoding design

### ➔ Express Values



### ➔ Separate, Order, Align Regions

➔ Separate



➔ Order



## Spatial orientation of axes

### ➔ Axis Orientation

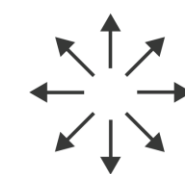
➔ Rectilinear



➔ Parallel

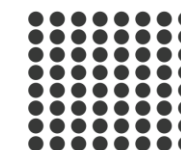


➔ Radial



### ➔ Layout Density

➔ Dense



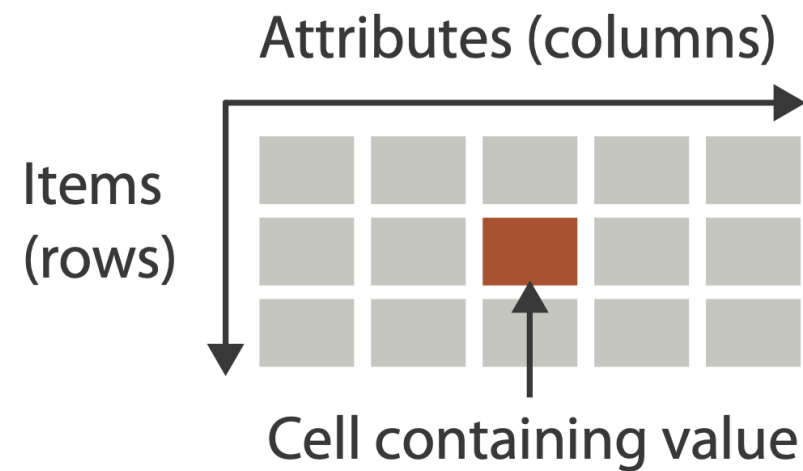
➔ Space-Filling



- Design choice - all aspects of the use of spatial channels for visual encoding.
- The use of space dominates the user's mental model of the dataset.
- The highest ranked **effectiveness** channel for categorical attributes, grouping items within the same region, is also about the use of space.

*The **effectiveness** principle dictates that the importance of the attribute should match the salience of the channel; that is, its noticeability*

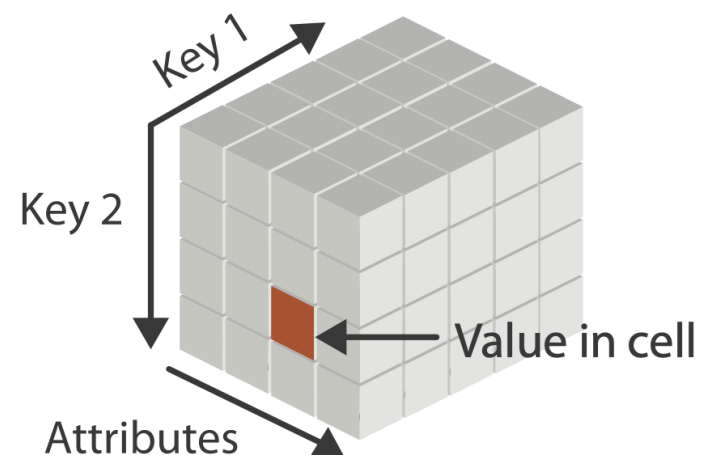
## → Tables



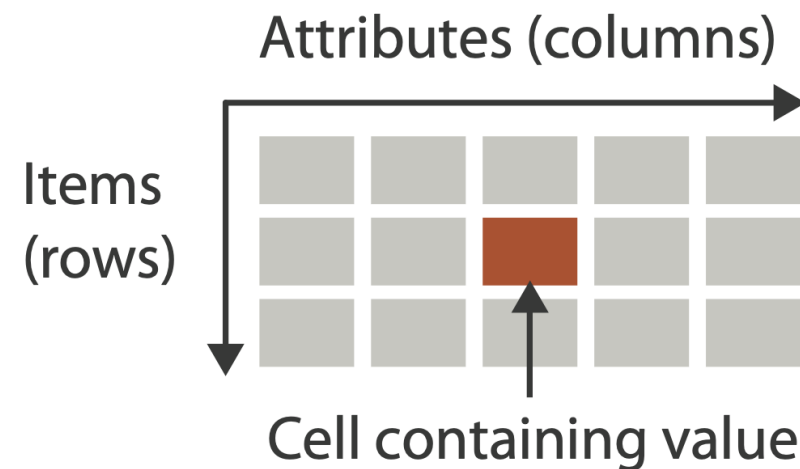
## KEYS

- **Independent** attribute
- Used as unique index to look up items
- Simple tables: **one key**
- Multidimensional tables: **multiple keys**

## → *Multidimensional Table*



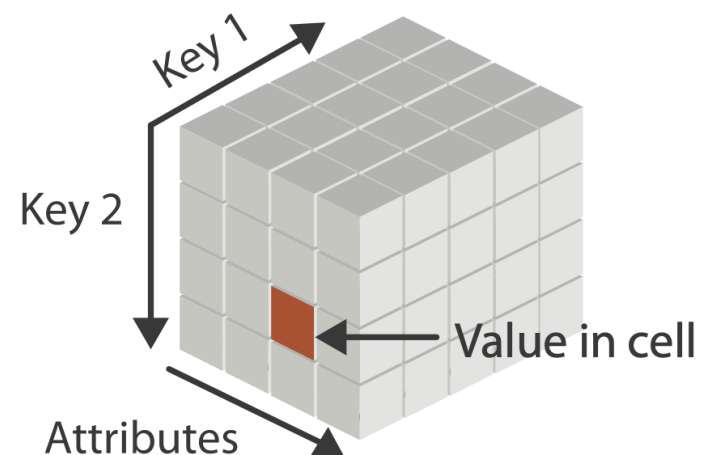
## → Tables



## VALUES

- **Dependent** attribute, value of cell
- Classify arrangements by key count
- **Zero, one, two, many...**

## → *Multidimensional Table*



## CLASSIFY ARRANGEMENTS BY KEYS USED

- 0, 1, 2, (keys)

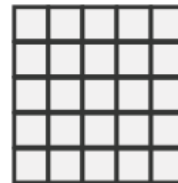
# Zero Keys



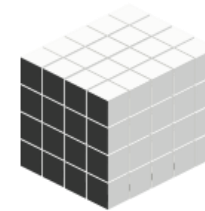
→ 1 Key  
*List*



→ 2 Keys  
*Matrix*



→ 3 Keys  
*Volume*



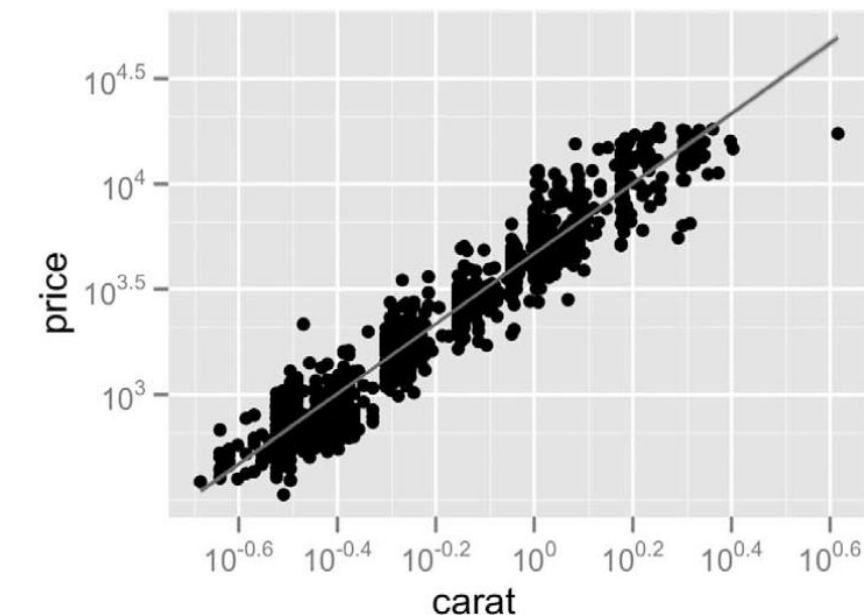
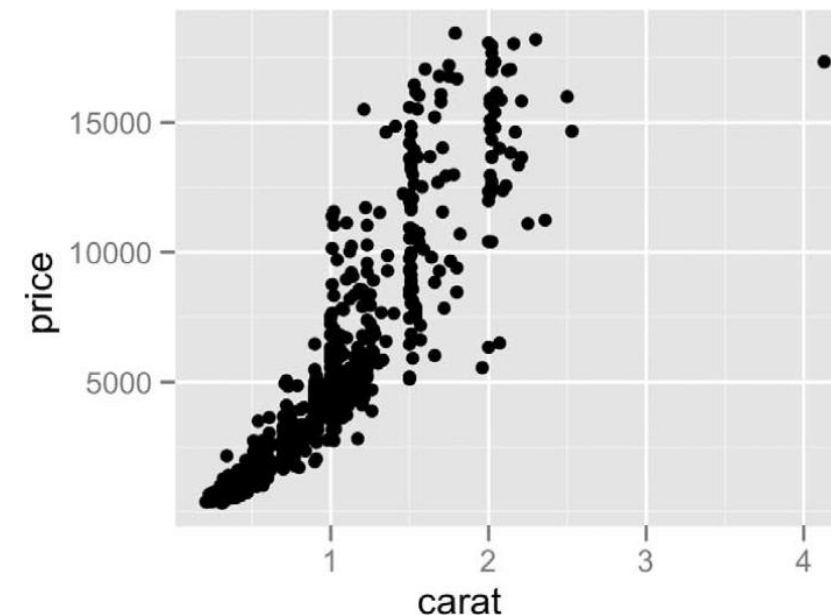
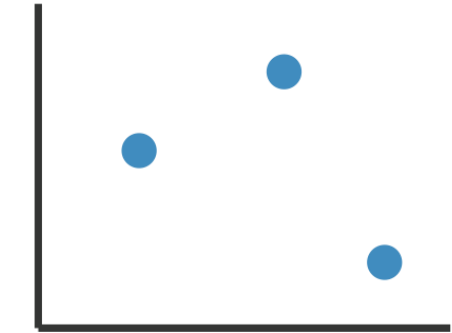
→ Many Keys  
*Recursive Subdivision*



# Zero Keys e.g., - Idiom: Scatterplot

- Express values
  - Quantitative attributes
- No keys, only values
  - **Data**: two quantitative attributes
  - **Mark**: points
  - **Channels**: horizontal and vertical position
  - **Tasks**: identify trends, outliers, distribution, correlation, clusters
  - **Scalability**: hundreds of items

## ➔ Express Values



[A layered grammar of graphics. Wickham. Journ. Computational and Graphical Statistics 19:1 (2010), 3–28.]

# Some Keys

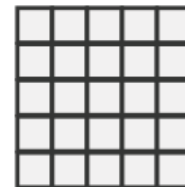
## → Express Values



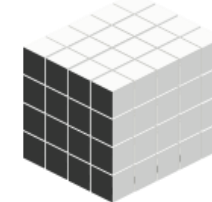
→ 1 Key  
*List*



→ 2 Keys  
*Matrix*



→ 3 Keys  
*Volume*



→ Many Keys  
*Recursive Subdivision*

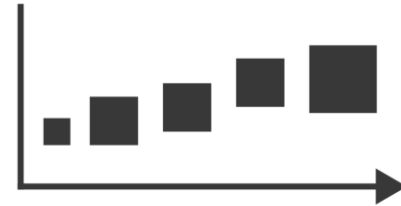


# Some Keys: Categorical Regions

→ Separate



→ Order



→ Align

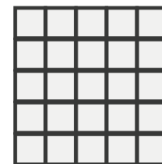


- **Regions:** contiguous bounded areas distinct from each other
  - Using space to **separate** (proximity)
  - Following expressiveness principle for categorical attributes
- Use ordered attribute to **order** and **align** regions

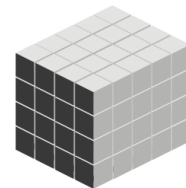
→ 1 Key  
*List*



→ 2 Keys  
*Matrix*



→ 3 Keys  
*Volume*



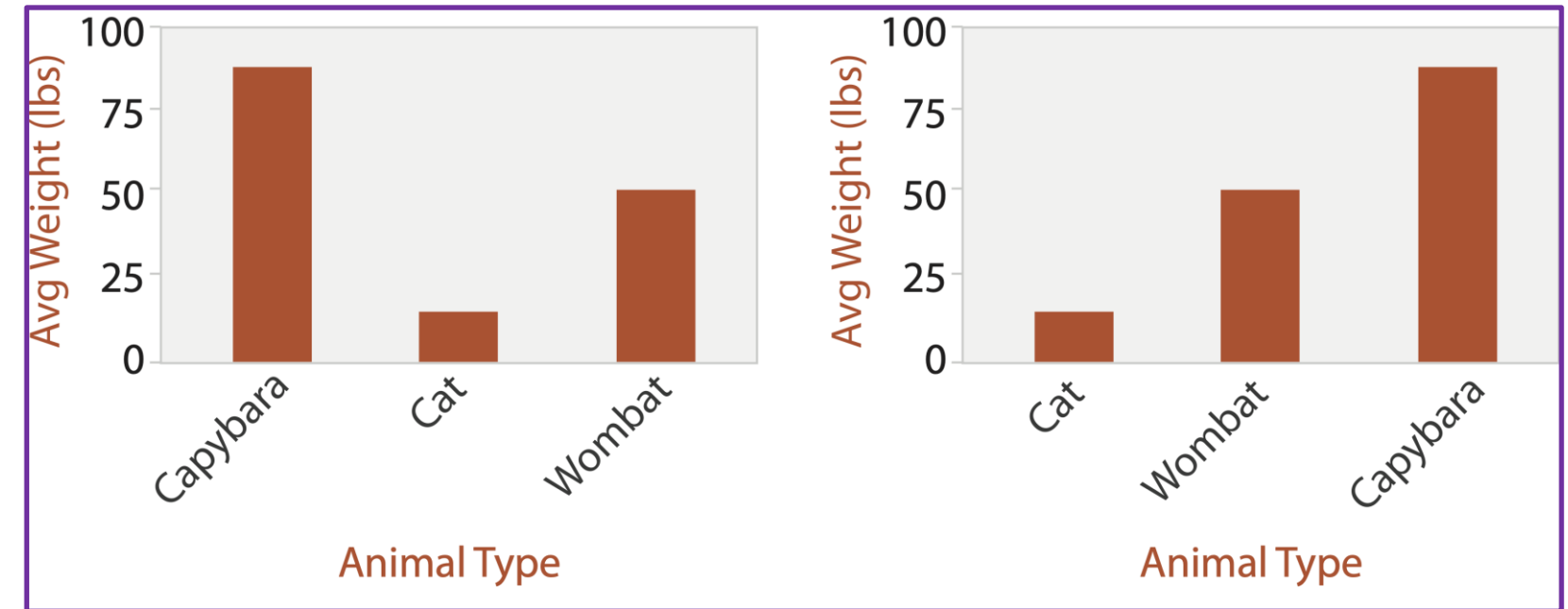
→ Many Keys  
*Recursive Subdivision*



The expressiveness principle dictates that the visual encoding should express all of, and only, the information in the dataset attributes.

# Idiom: Bar chart

- One key, one value
  - **Data:** one categorical attribute, one quantitative attribute
  - **Mark:** lines
  - **Channels:**
    - Length to express quantitative value
    - Spatial regions: one per mark
      - separated horizontally, aligned vertically
      - ordered by quantitative attribute
        - by label (alphabetical)
        - by length attribute (data-driven)
  - **Tasks:** compare, lookup values
  - **Scalability:** dozens to hundreds of levels for key attribute



The value attribute, weight, expresses the value with aligned vertical spatial position and line marks.

- (a) Marks ordered alphabetically according to species name.
- (b) Marks ordered by the weight attribute used for bar heights.

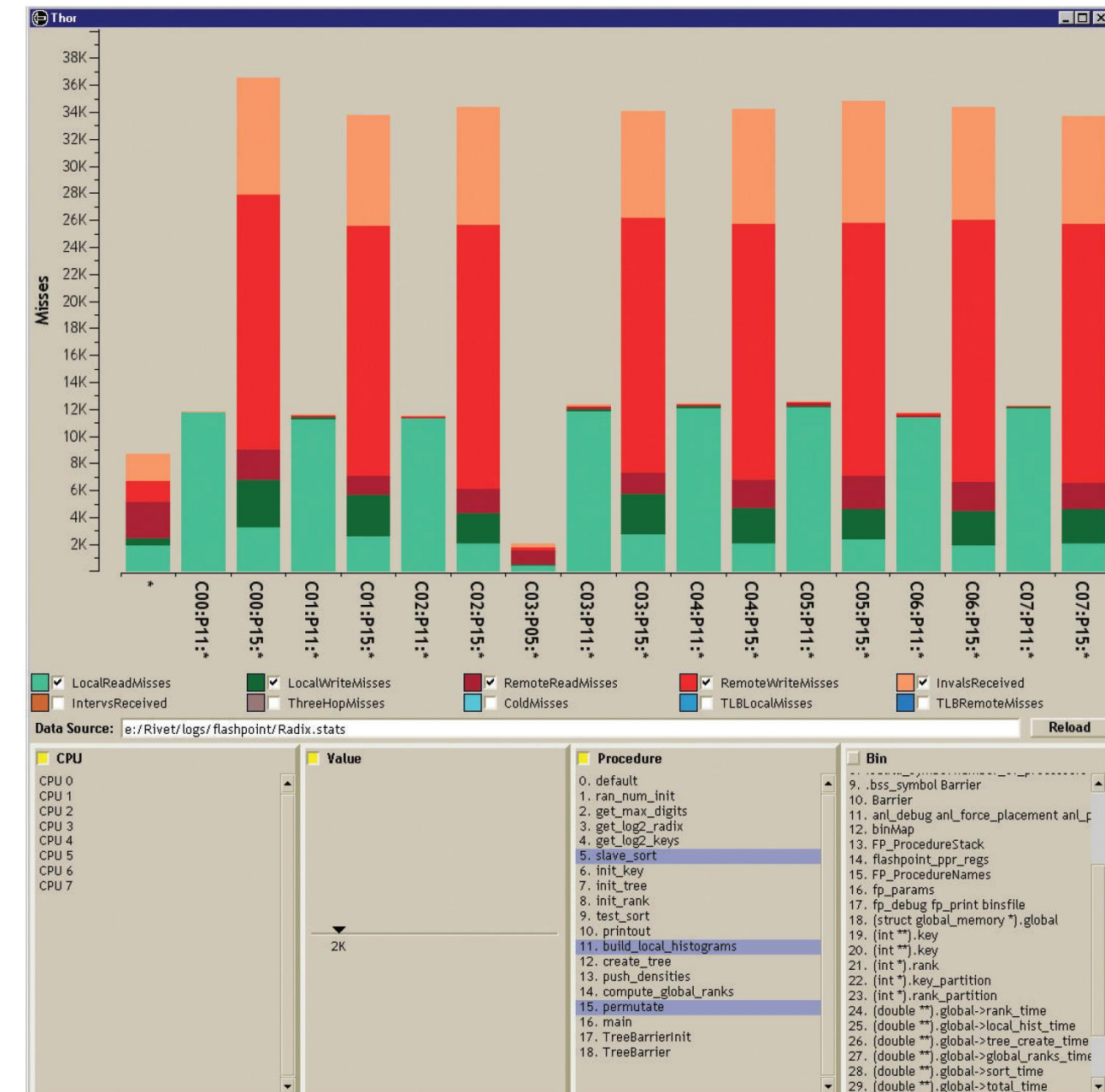
## Example:

<https://observablehq.com/@berkeleyvis/separate-order-and-align>

- **Separate:** helps differentiating
- **Order:** helps locating/lookup
- **Align:** helps comparing

# Idiom: Stacked Bar Chart

- One more key (primary key – arranged horizontally, secondary key – vertical structure of glyphs)
  - **Data:** 2 category attributes, 1 quantitative attribute
  - **Mark:** vertical stack of line marks
    - glyph: composite object, internal structure from multiple marks
  - **Channels**
    - length and colour hue
    - spatial regions: one per glyph
      - aligned: full glyph, lowest bar component
      - unaligned: other bar components
  - **Task:** part-to-whole relationship (subcomponent within a stacked bar), lookup values, find trends.
  - **Scalability:** dozens to hundreds of levels for primary key attributes and several to one dozen levels for stacked attributes



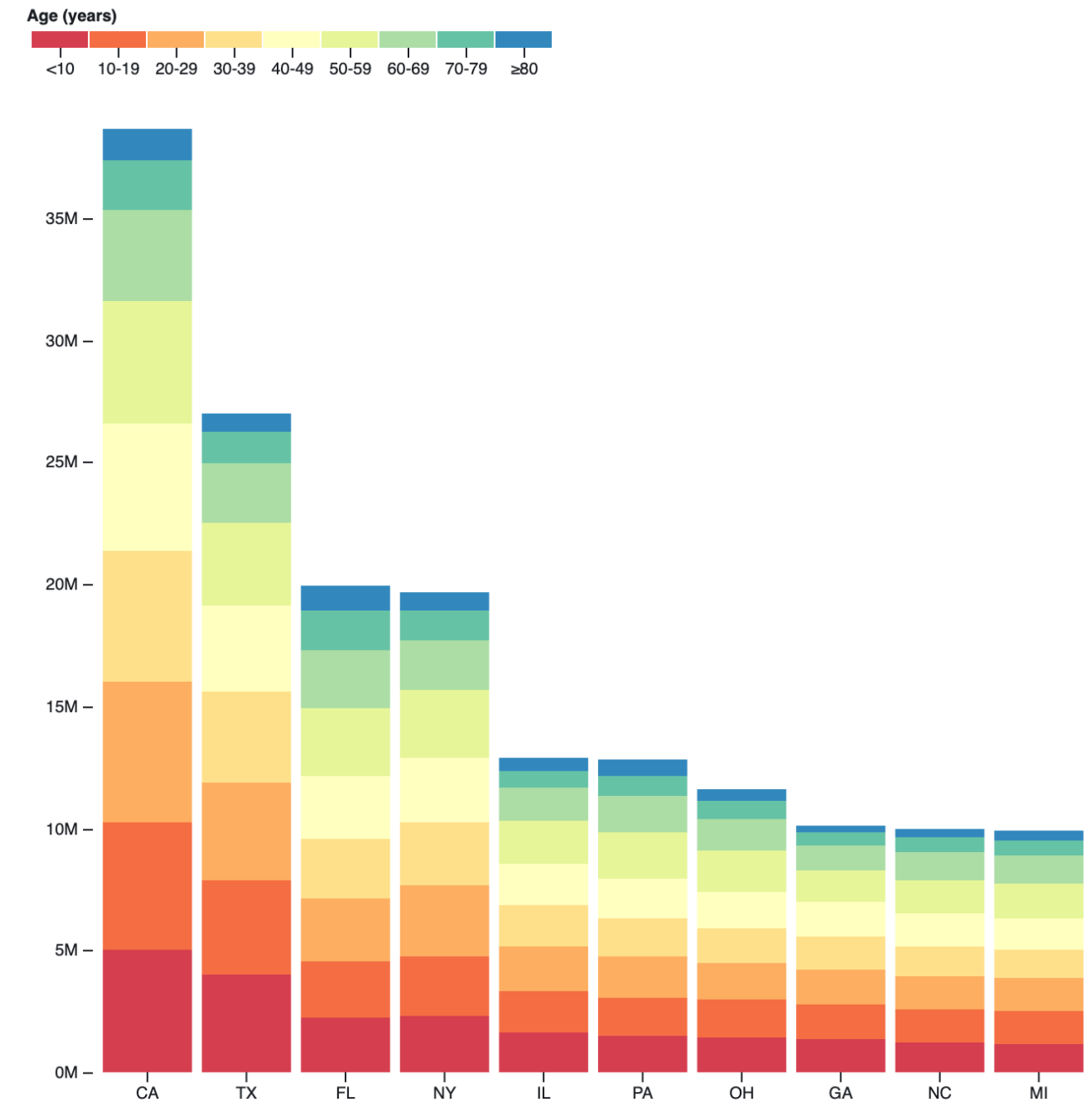
Computer memory profiler. From [Bosch 01]. The type of cache miss (colour); the height of each full bar (glyphs) encodes all cache misses for each processor–procedure combination

# Idiom: Stacked Bar Chart

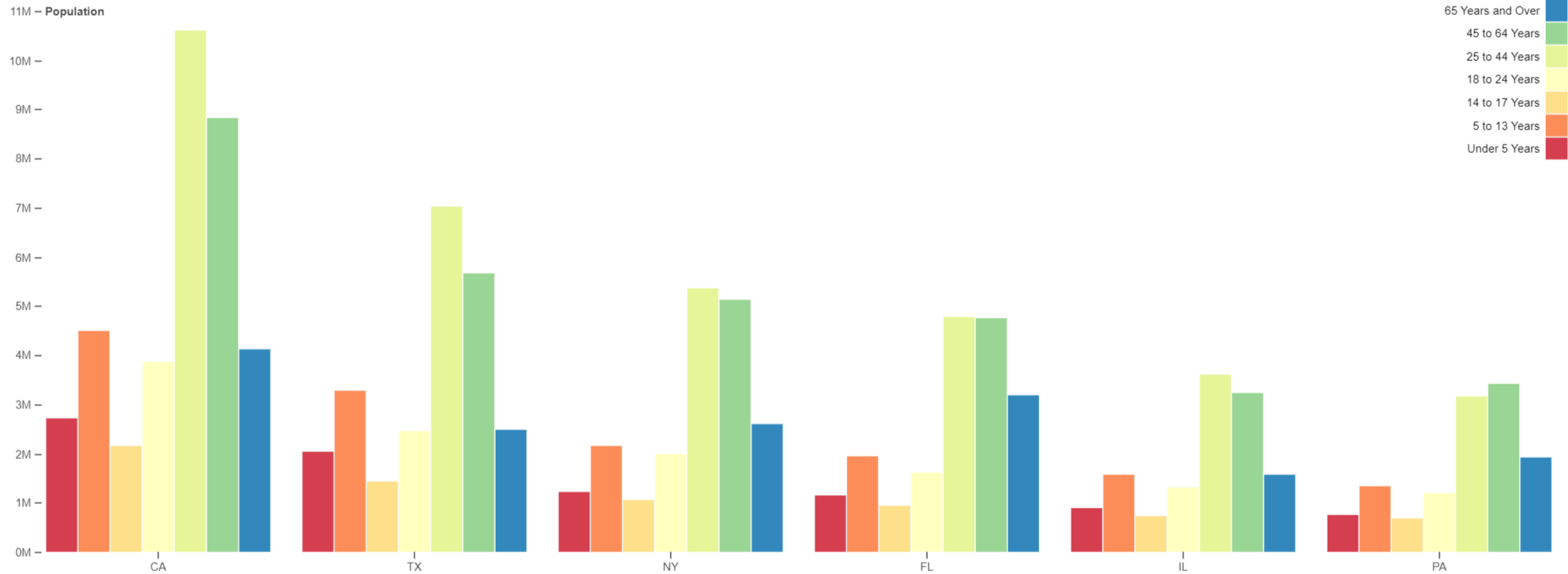
- Other example

## Stacked Bar Chart

Population by age and state. Compare to [horizontal stacked bars](#), [normalized stacked bars](#) and a [dot plot](#). Data: [American Community Survey](#)

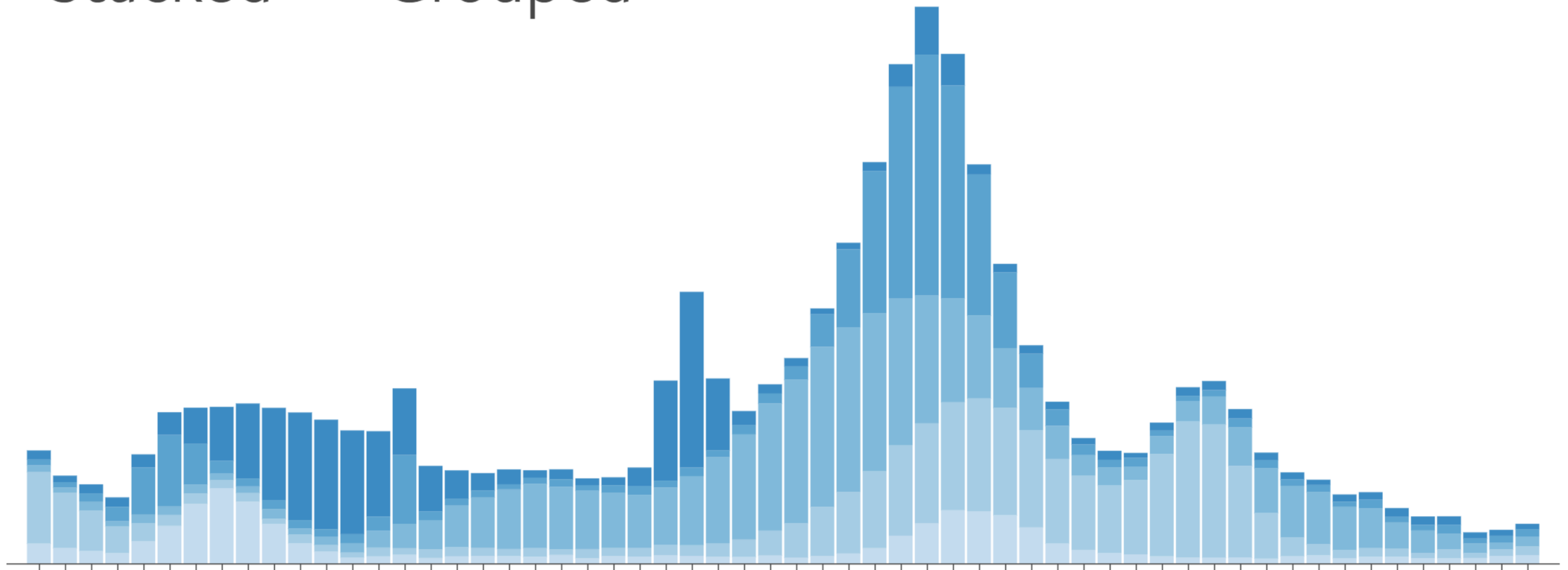


# Grouped Bar Chart



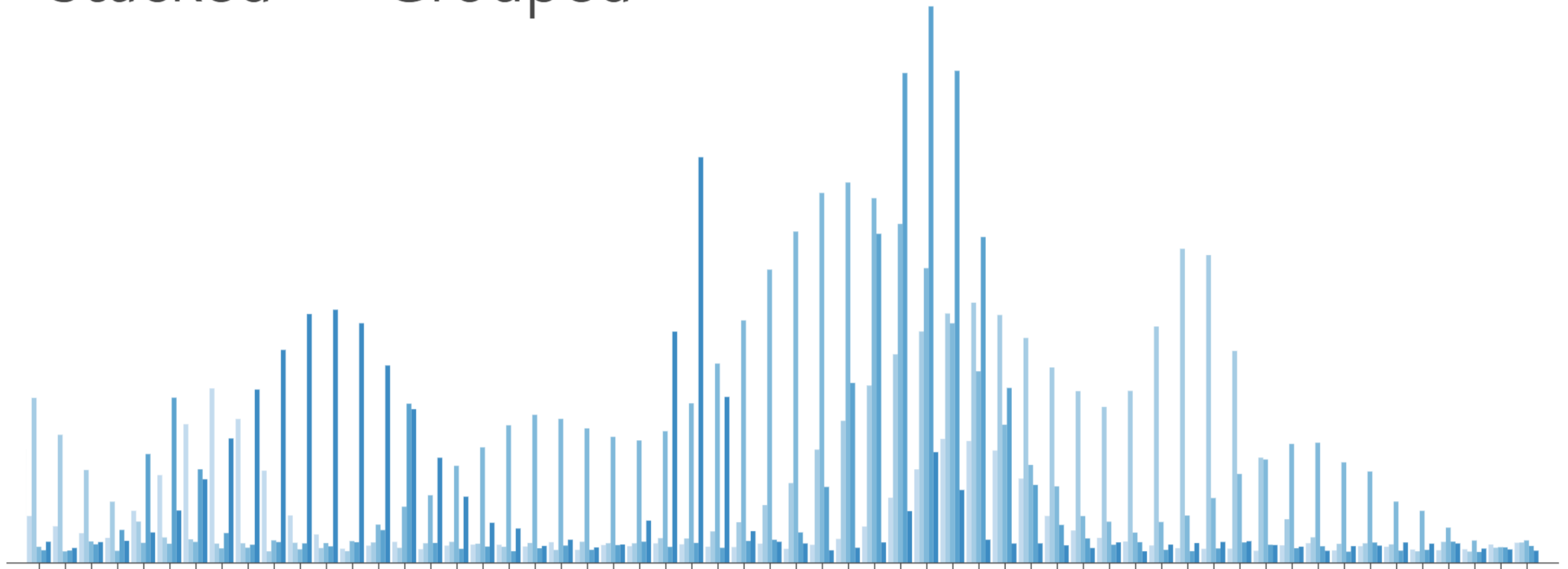
# Stacked to Grouped

- Stacked
- Grouped



# Stacked to Grouped

- Stacked
- Grouped



## Stacked bar charts are the worst

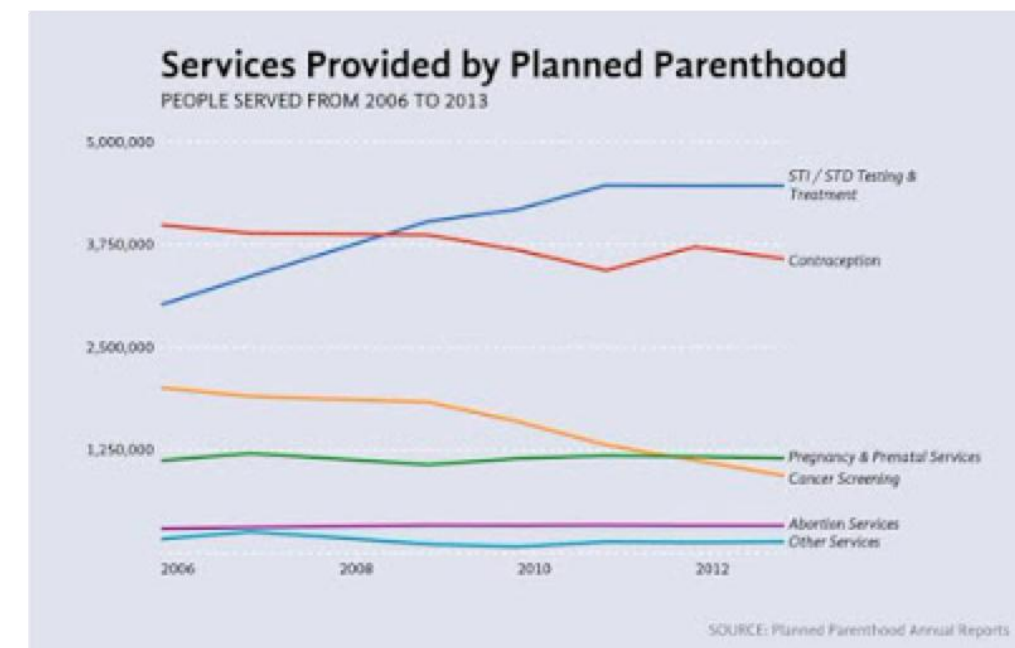
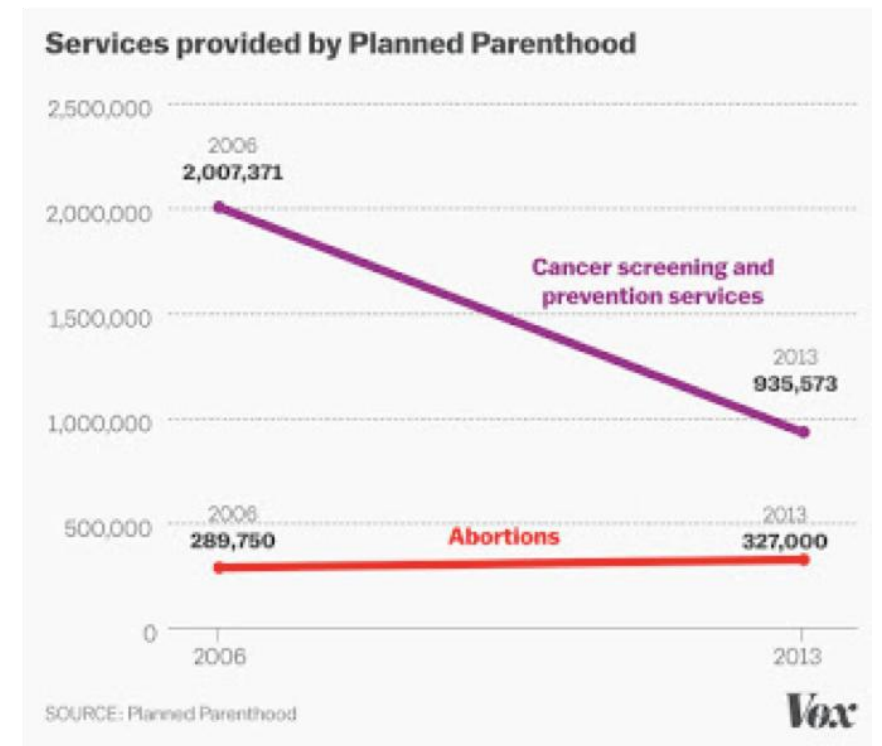
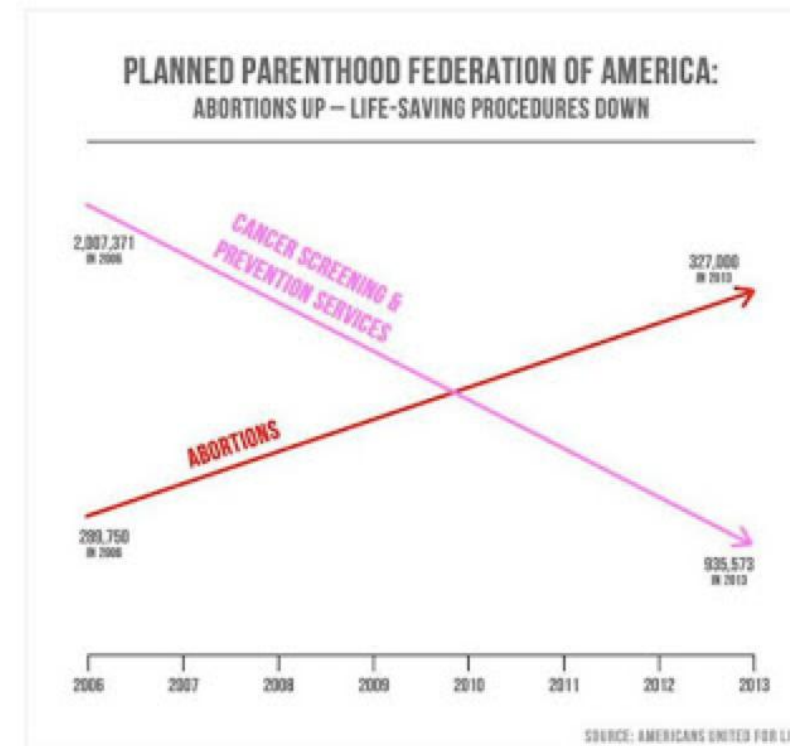
Interesting discussion from Robert Kosara's blog.

<https://eagereyes.org/techniques/stacked-bars-are-the-worst>

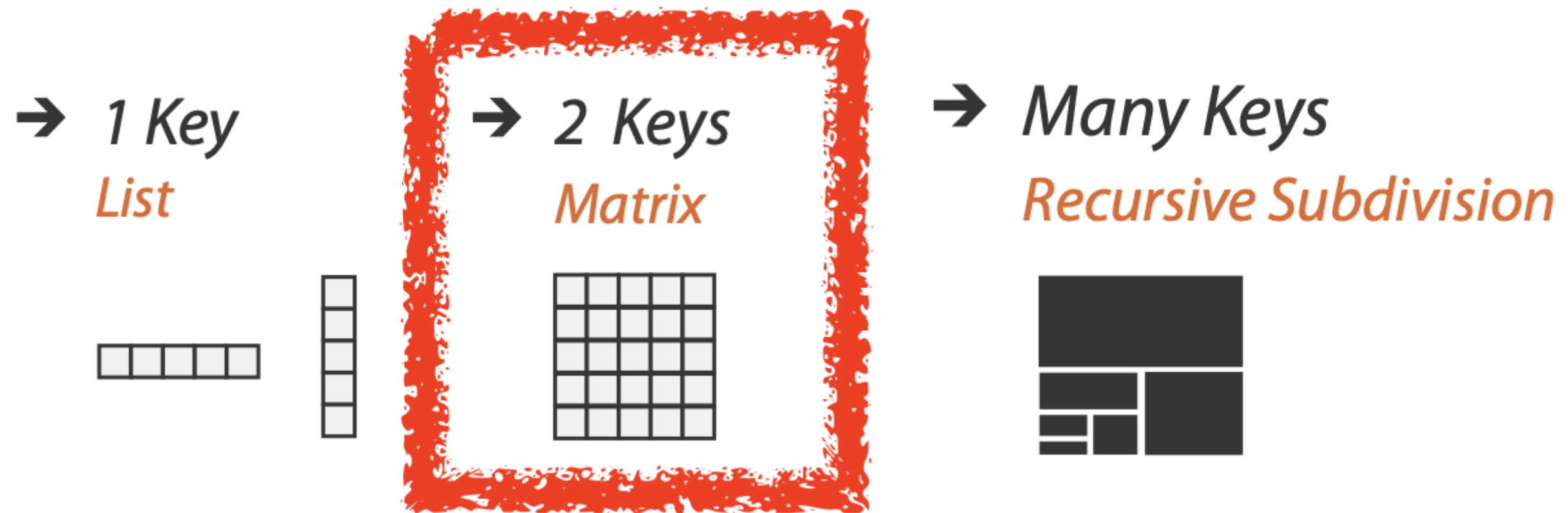
*Make sure to read the comments.*

# Chart Axes

- Labelled axis is critical
- Avoid cropping y-axis
- Include 0 at bottom left
- Avoid slope misleads



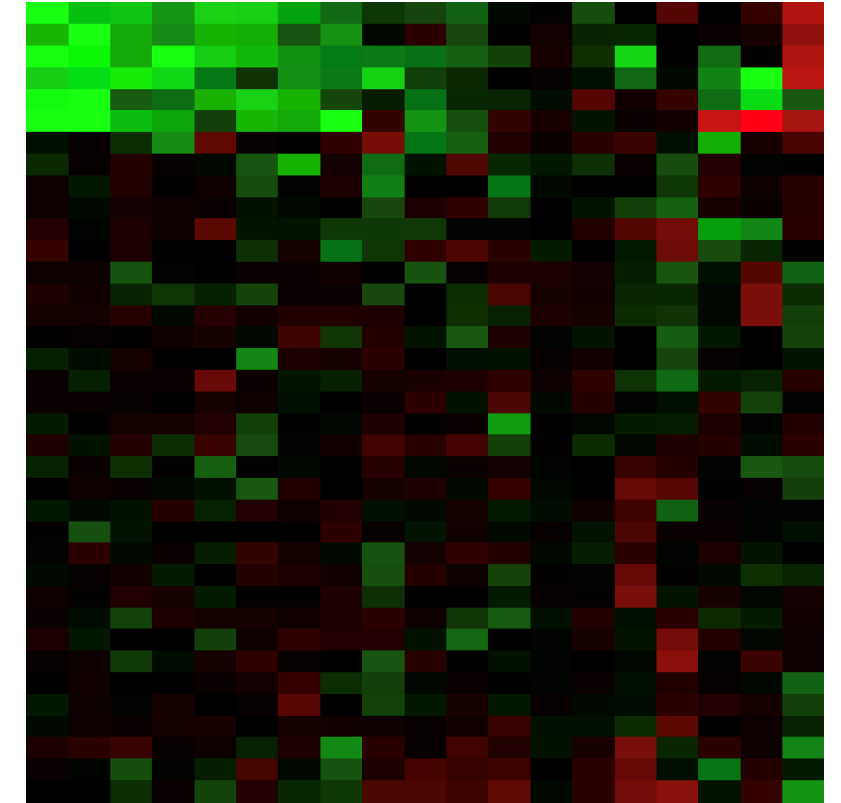
# TWO KEYS



# Idiom: Heatmap (Two Keys)

Two keys, one value

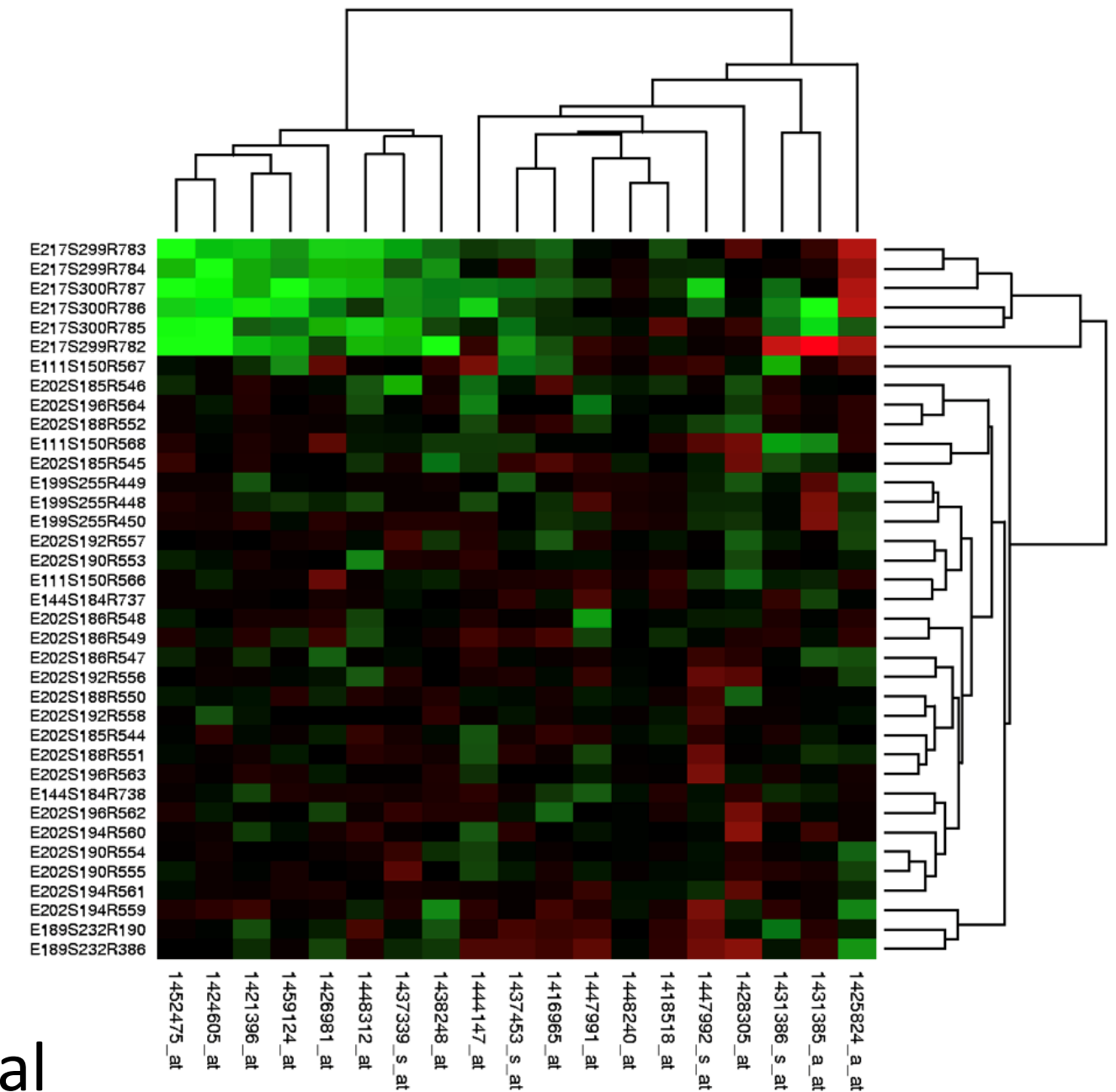
- **Data:**
  - 2 category attributes (gene, experimental condition)
  - 1 quantity attribute (expression levels)
- **Marks:** area
  - separate and align in 2D matrix
  - indexed by 2 categorical attributes
- **Channels:**
  - colour by quantitative attribute (ordered diverging colour map, e.g., red to green)
- **Task:** find clusters, outliers, summarise
- **Scalability:**
  - 1M items, 100s of categorical levels (200X200 (a single pixel), ~10 quantitative attribute levels)
- Small area mark is very compact, good for overview, with high information density



# Idiom: Cluster Heatmap

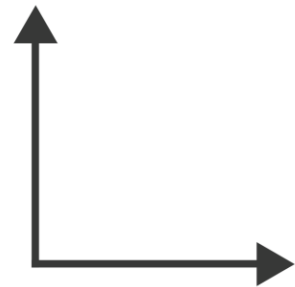
In addition

- Derived data
  - 2 cluster hierarchies
- Dendrogram
  - Parent-child relationships in tree with connection line marks
  - Leaves aligned so interior branch heights easy to compare
- Heatmap
  - Marks (re-)ordered by cluster hierarchy traversal
  - Task: assess quality of clusters found by automatic methods

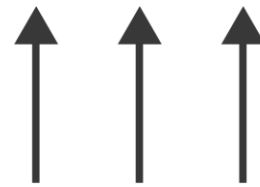


## ➔ Axis Orientation

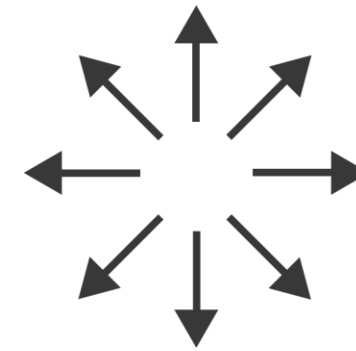
➔ Rectilinear



➔ Parallel



➔ Radial



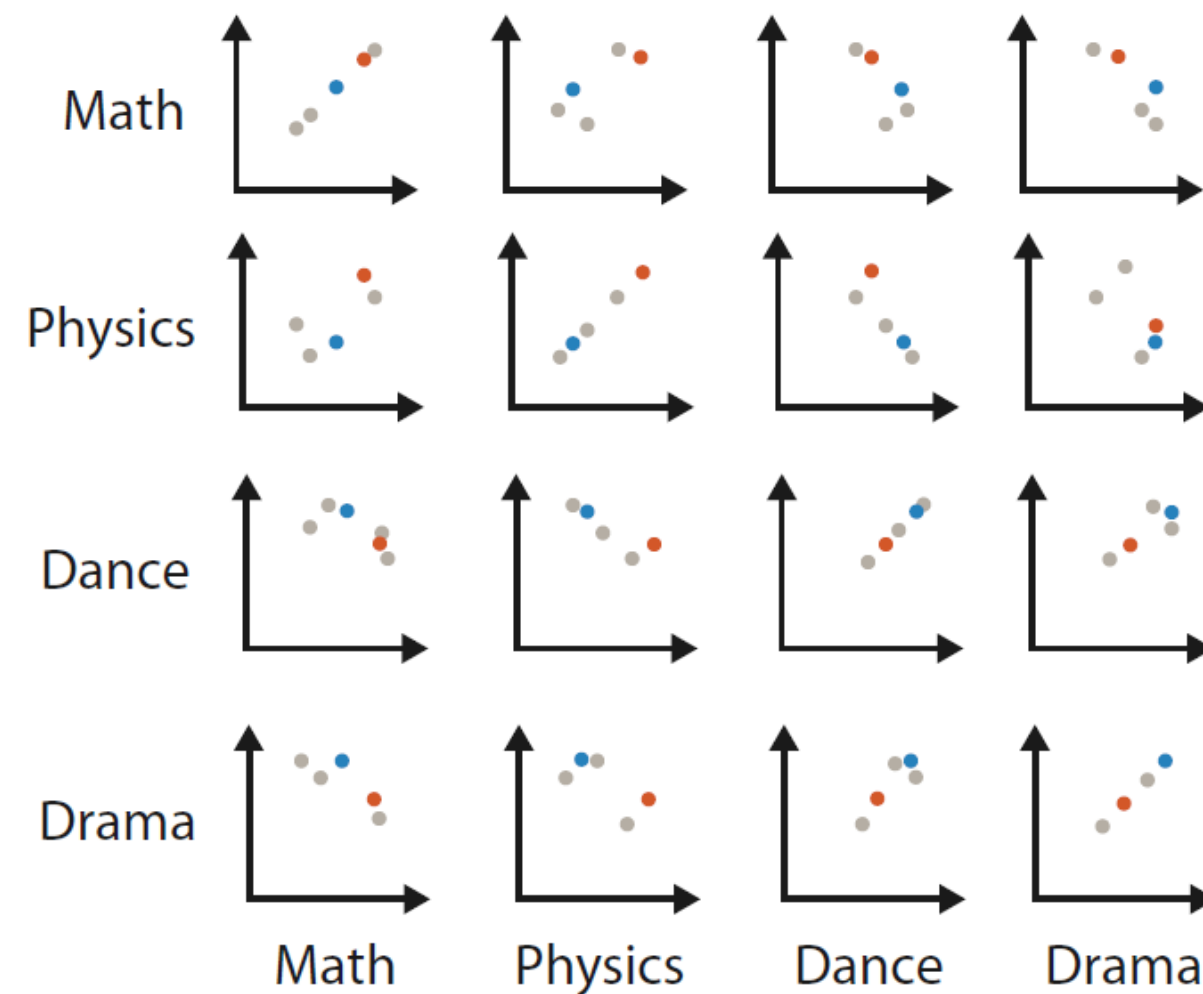
# Idiom: Scatterplot Matrix

Data:

Table

| Math | Physics | Dance | Drama |
|------|---------|-------|-------|
| 85   | 95      | 70    | 65    |
| 90   | 80      | 60    | 50    |
| 65   | 50      | 90    | 90    |
| 50   | 40      | 95    | 80    |
| 40   | 60      | 80    | 90    |

Scatterplot Matrix



## Idiom - Scatterplot Matrix (SPLOM)

- Rectilinear axes, point mark
- All possible pairs of axes
- Scalability
  - one dozen attributes
  - dozens to hundreds of items
- Task: summarize/explore

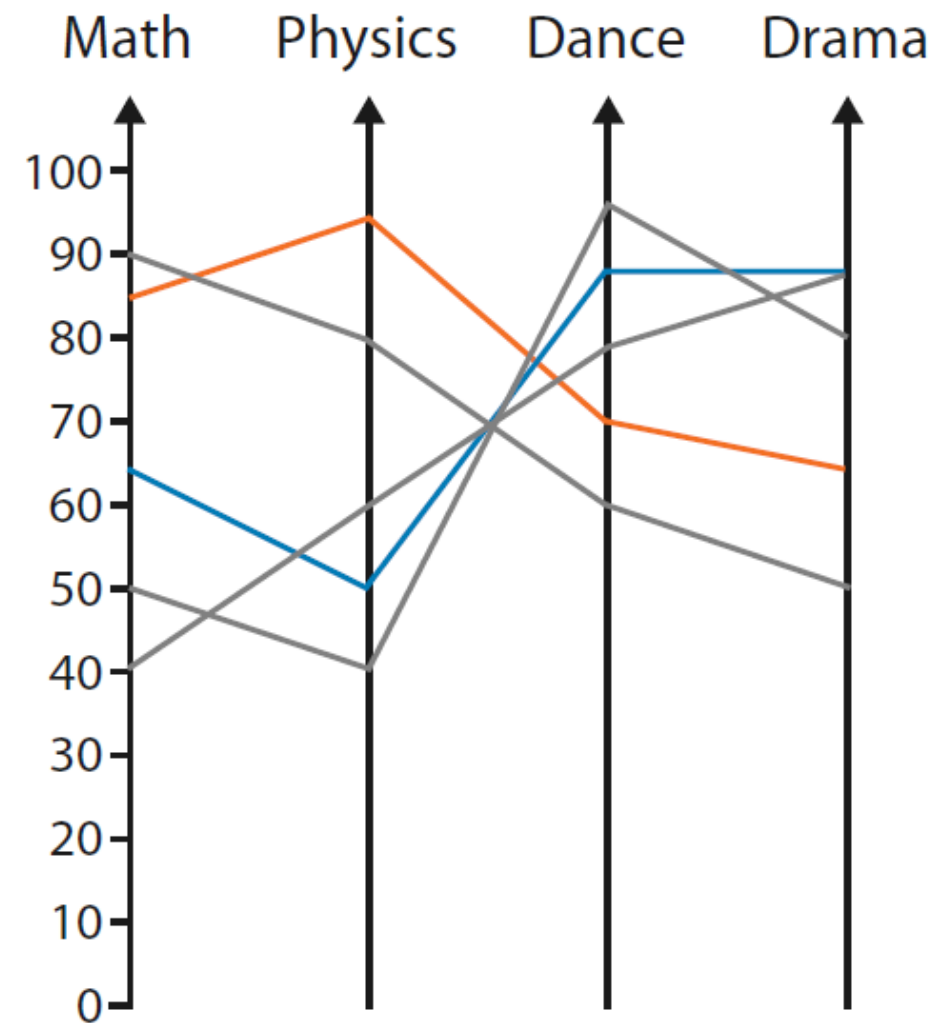
# Idiom: Parallel Coordinates

Data:

Table

| Math | Physics | Dance | Drama |
|------|---------|-------|-------|
| 85   | 95      | 70    | 65    |
| 90   | 80      | 60    | 50    |
| 65   | 50      | 90    | 90    |
| 50   | 40      | 95    | 80    |
| 40   | 60      | 80    | 90    |

Parallel Coordinates



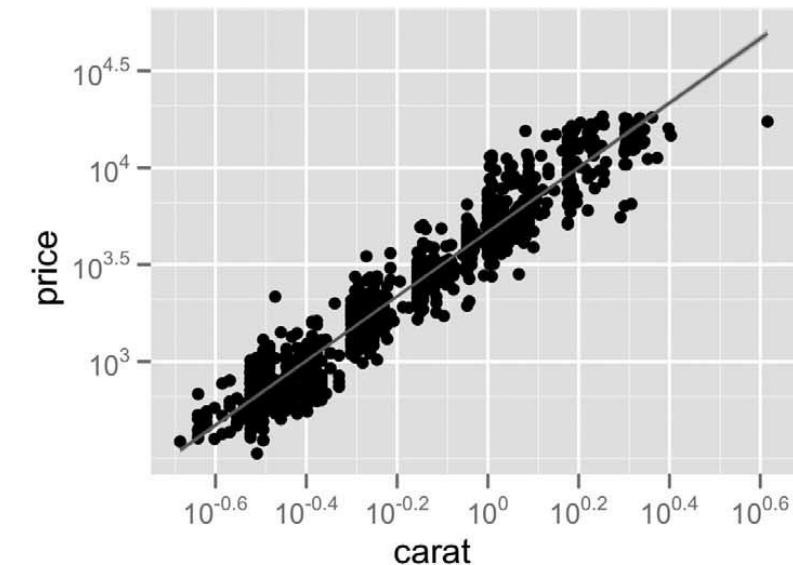
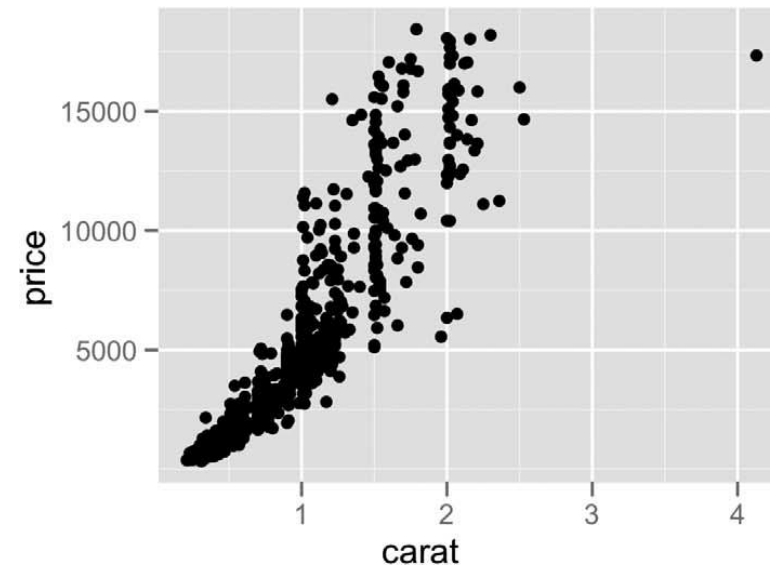
Idiom - **Parallel Coordinates**

- Parallel axes, jagged line representing item
- Rectilinear axes, item as point
  - axis ordering is major challenge
- Scalability
  - dozens of attributes
  - hundreds of items
- Task: summarize/explore

# Task Correlation

- **Scatterplot matrix**

- Positive correlation
  - diagonal low-to-high
- Negative correlation
  - diagonal high-to-low
- Uncorrelated: spread out



- **Parallel coordinates**

- Positive correlation
  - parallel line segments
- Negative correlation
  - all segments cross at halfway point
- Uncorrelated
  - scattered crossings

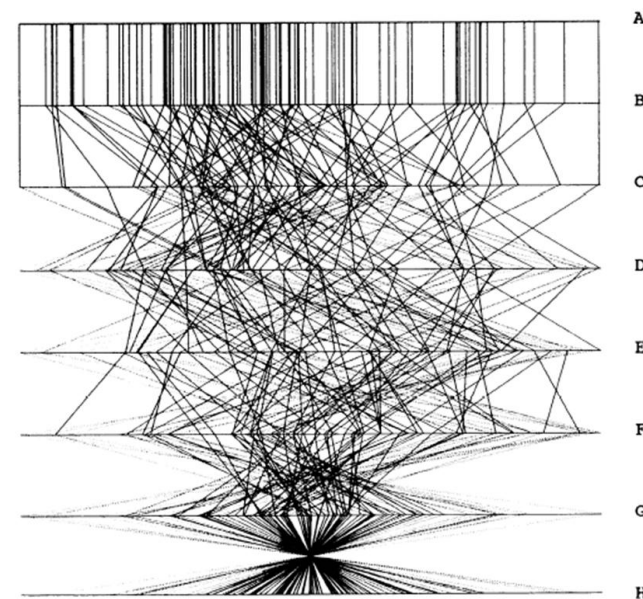
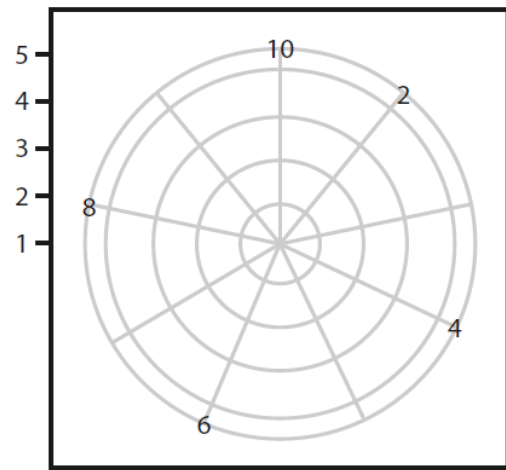


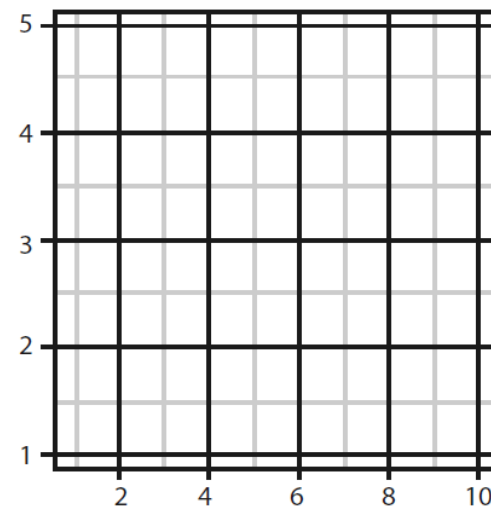
Figure 3. Parallel Coordinate Plot of Six-Dimensional Data Illustrating Correlations of  $\rho = 1, .8, .2, 0, -.2, -.8, \text{ and } -1$ .

*Parallel coordinates were designed to show correlation between neighbouring axes. At the **top**, parallel lines show perfect positive correlation. At the **bottom**, all of the lines cross over each other at a single spot in between the two axes, showing perfect negative correlation. In the **middle**, the mix of crossings shows uncorrelated data. From [Wegman 90, Figure 3].*

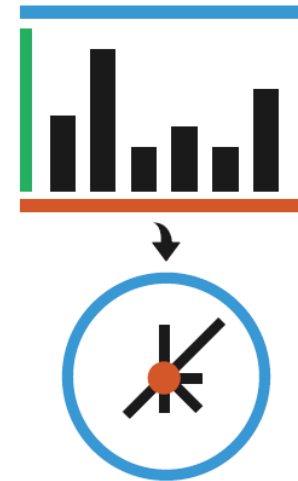
# Spatial Axis Orientation: Radial Layouts



(a)



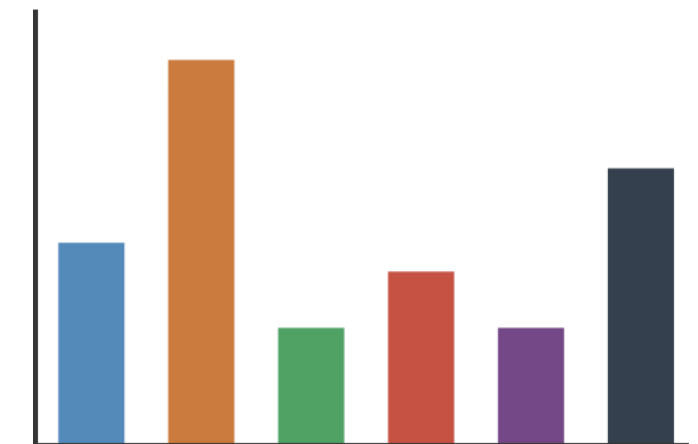
(b)



(c)

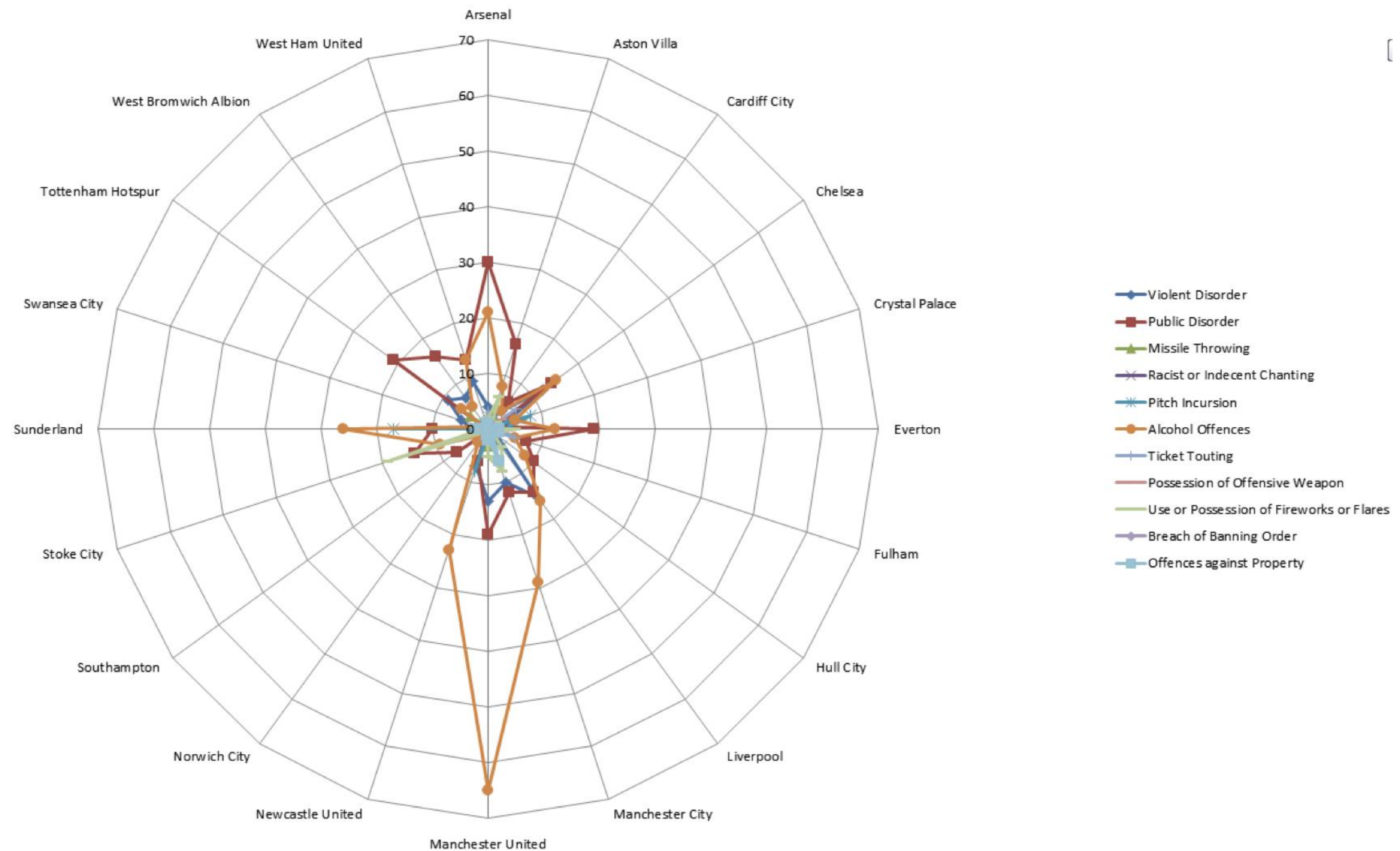
Layout coordinate systems.

- (a) Radial layouts use polar coordinates, with one spatial position and one angle channel.
- (b) Rectilinear layouts use two perpendicular spatial position channels. After [Wickham 10].
- (c) Transforming rectilinear to radial layouts maps two parallel bounding lines to a point at the centre and a circle at the perimeter.

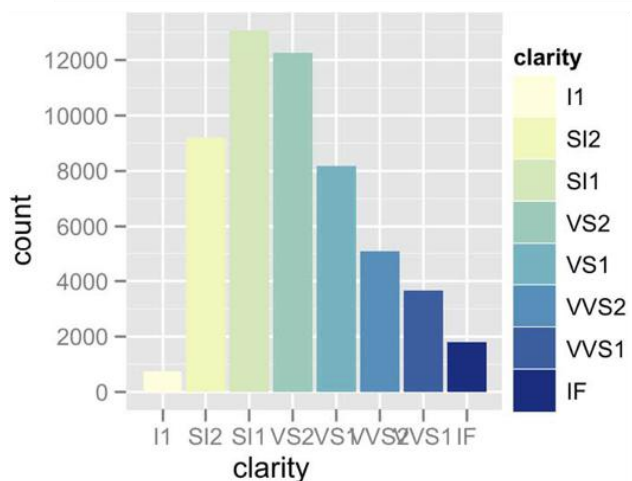
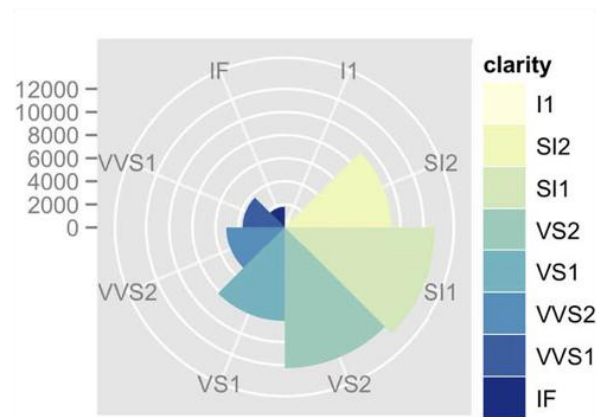
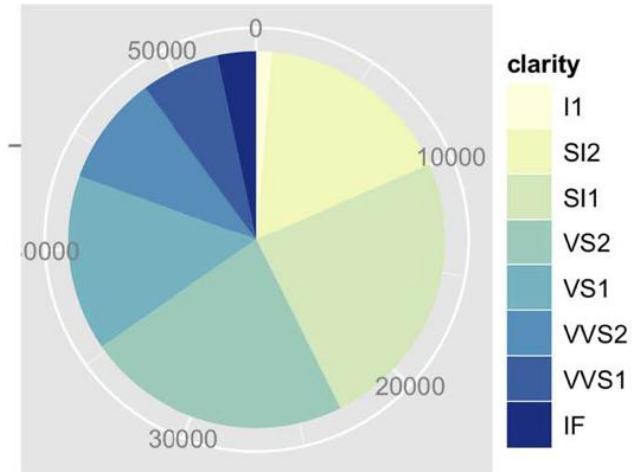


# Idiom: Radar Plots

- Task: comparison
- Scalability: similar to parallel coordinates
- Order of attributes matter

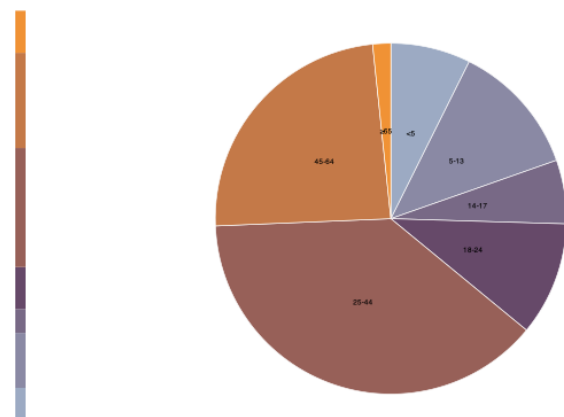
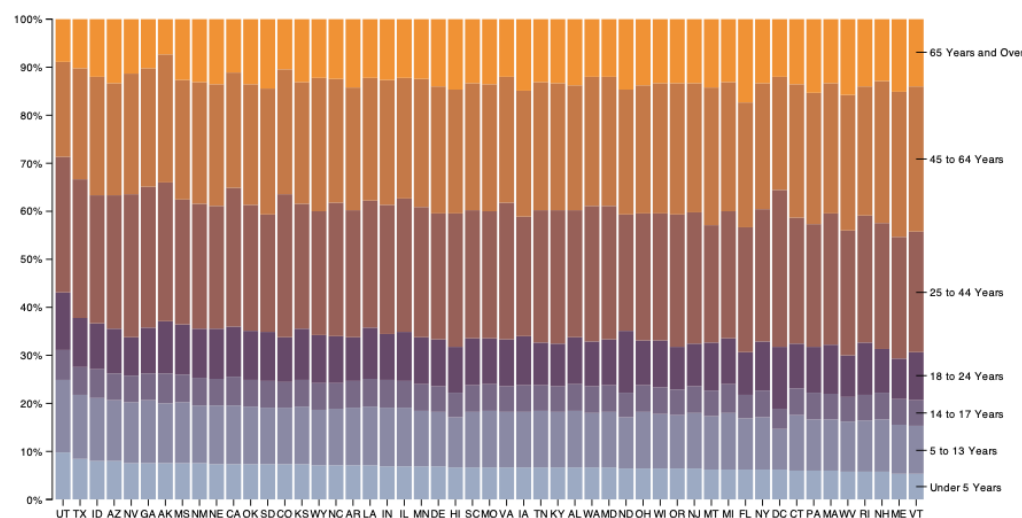
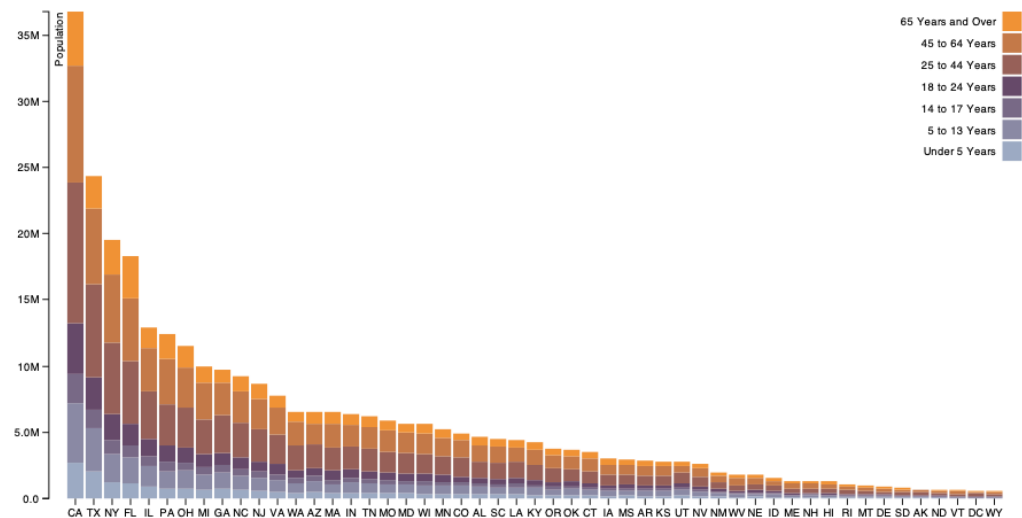


# Idioms: Pie Chart, Polar Area Charts



- Pie chart
  - Area marks with angle channel
  - Accuracy: angle/area much less accurate than line length
- Polar area chart
  - Area marks with length channel
  - More direct analog to bar charts
- Data
  - 1 categorical key attribute, 1 quantitative value attribute
- Task
  - part-to-whole judgements

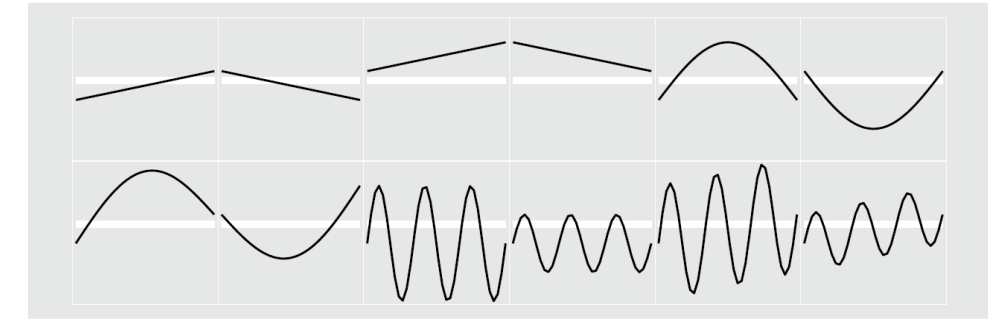
# Idioms: Normalized Stacked Bar Chart



- Task: part-to-whole judgements
- Normalized stacked bar chart
  - stacked bar chart, normalized to full vertical height
  - single stacked bar equivalent to full pie
    - high information density: requires narrow rectangle
- Idiom - pie chart:
  - information density: requires large circle

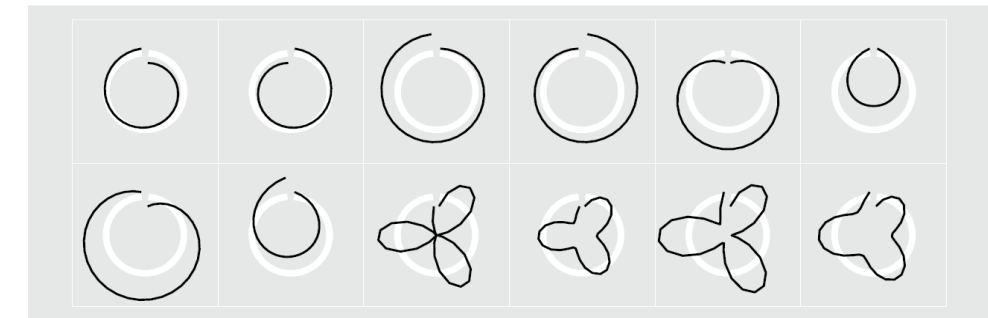
# Idiom: Glyphmaps

**(a) rectilinear** good for linear vs nonlinear trends



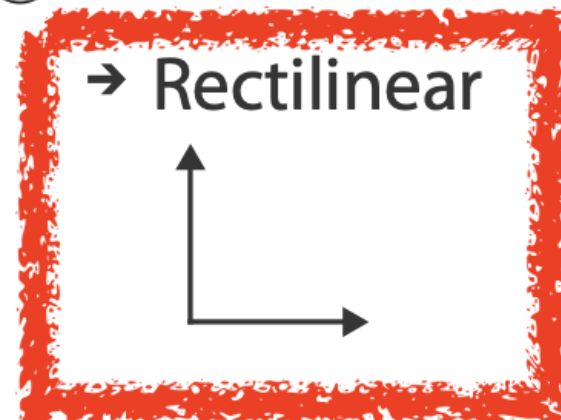
(a)

**(b) radial** good for cyclic patterns



(b)

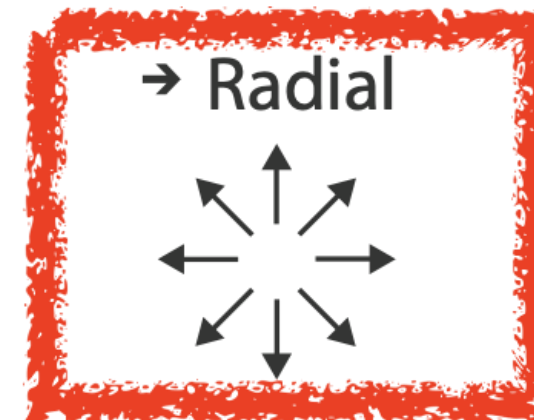
➔ **Axis Orientation**



➔ Parallel



➔ Radial



- **Rectilinear:** scalability with regard to #axes

- Two axes best
- Three problematic
- Four or more **impossible**

- **Parallel:** unfamiliarity, training time

- **Radial:** perceptual limits

- Angles lower precision than length

➞ Axis Orientation

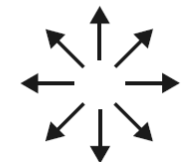
➞ Rectilinear



➞ Parallel

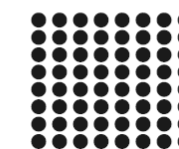


➞ Radial



➞ Layout Density

➞ Dense



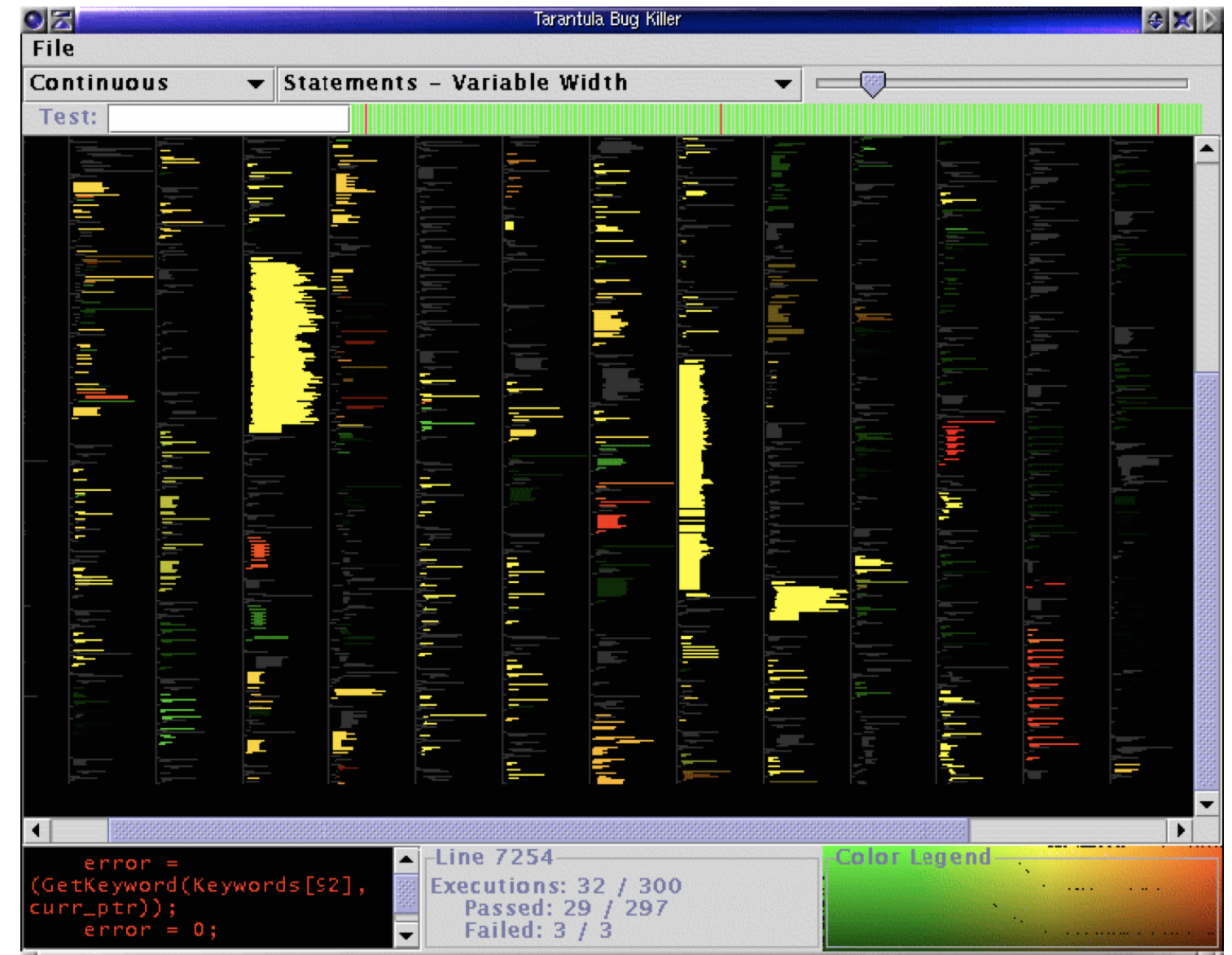
➞ Space-Filling



# Idiom: Dense Overview

A dense layout (also called pixel oriented) uses small and densely packed marks to provide an overview of as many items as possible with very high information density.

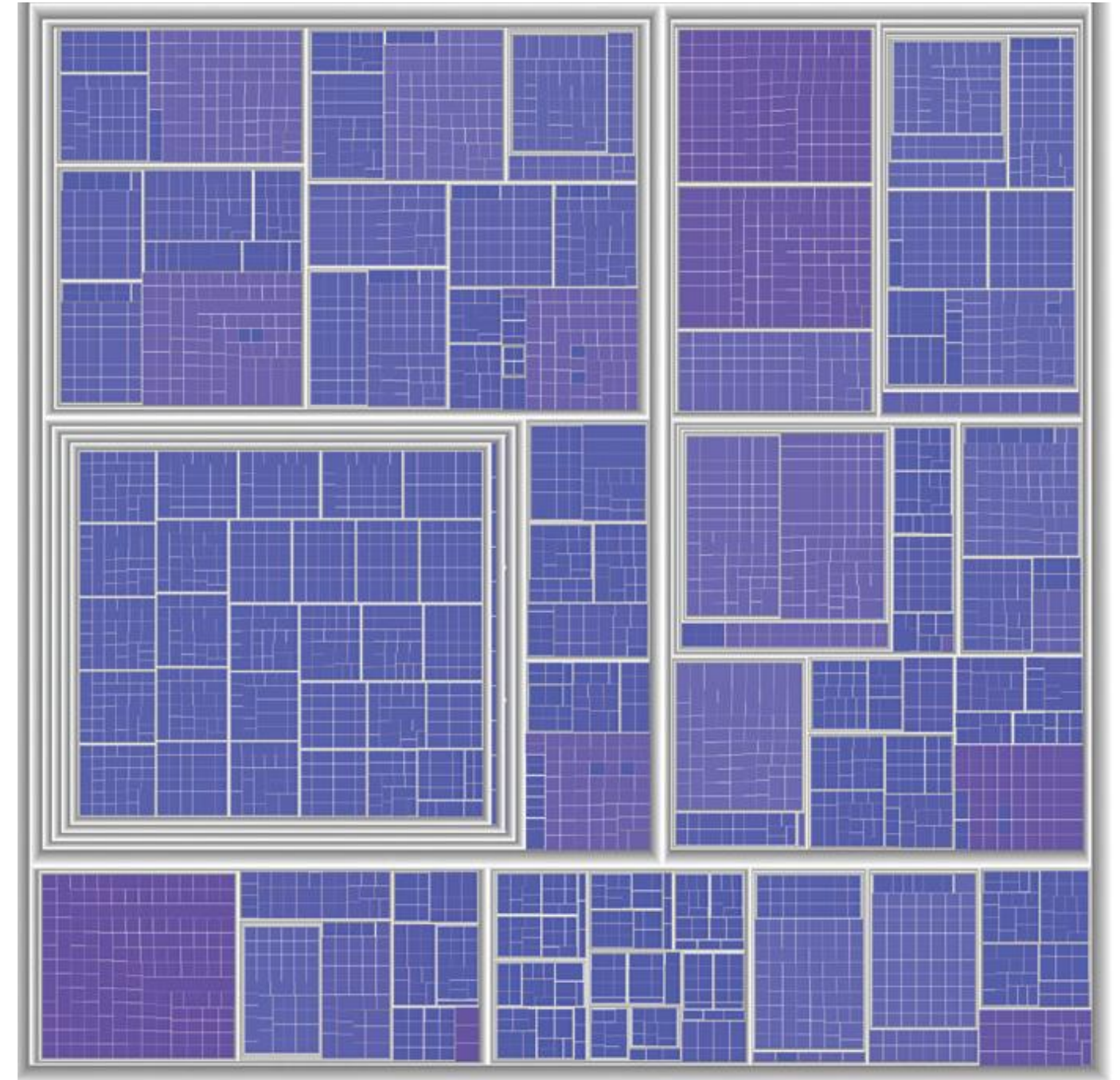
- Data: text
  - Text and one quantitative attribute per line
- Derived data:
  - One pixel high line
  - Length according to original
- Color line by attribute
- Scalability
  - Over 10,000 lines



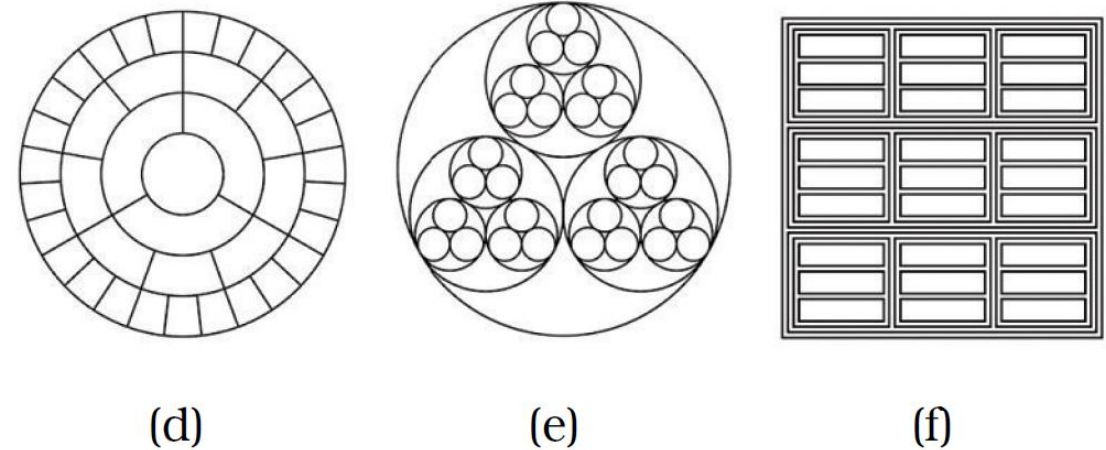
Tarantula shows a dense overview of source code with lines colour coded by execution status of a software test suite. From [Jones et al. 02].  
The arrangement of the marks is dictated by the order and length of the lines of code

# Idiom: Treemap

- Space-filling layout - fills all available space in the view, as the name implies.
- Space-filling layouts typically use area marks for items or containment marks for relationships, rather than line or connection marks, or point marks.



- Advantage – maximize the amount of room available for colour coding, increasing the chance that the coloured region will be large enough to be perceptually salient to the viewer.
- A related advantage - available space representing an item is often large enough to show a label embedded within it, rather than needing more room off to the side.



(d) **Concentric** circles, with radial depth spatial position and size showing tree depth and radial angular spatial position showing link relationships and sibling order.

(e) **Nested** circles, using radial containment, with nesting level and size showing tree depth.

(f) **Treemap**, using rectilinear containment, with nesting level and size showing tree depth.

*(Extracted from Figure 9.9)*

# Summary: Arrange Table

## ➔ Express Values



## ➔ Separate, Order, Align Regions

➔ Separate



➔ Order



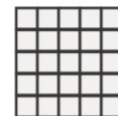
➔ Align



➔ 1 Key  
*List*



➔ 2 Keys  
*Matrix*



## ➔ Axis Orientation

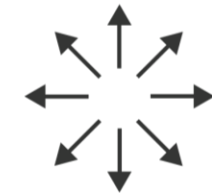
➔ Rectilinear



➔ Parallel

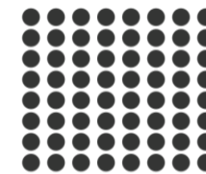


➔ Radial



## ➔ Layout Density

➔ Dense



➔ Space-Filling



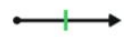
# Summary: Arrange Table

## How?

### Encode

#### ➔ Arrange

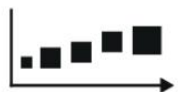
➔ Express



➔ Separate



➔ Order



➔ Align



➔ Use



#### ➔ Map

from **categorical** and **ordered** attributes

➔ Color

➔ Hue



➔ Saturation



➔ Luminance



➔ Size, Angle, Curvature, ...



➔ Shape



➔ Motion

Direction, Rate, Frequency, ...



### Manipulate

#### ➔ Change



#### ➔ Select

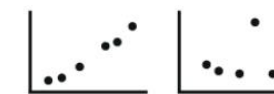


#### ➔ Navigate



### Facet

#### ➔ Juxtapose



#### ➔ Partition



#### ➔ Superimpose



### Reduce

#### ➔ Filter



#### ➔ Aggregate



#### ➔ Embed



What?

Why?

How?

# Further reading

- Visualization Analysis and Design. Munzner. AK Peters Visualization Series, CRC Press, Nov 2014.
  - Chap 7: Arrange Tables
- Visualizing Data. Cleveland. Hobart Press, 1993.
- A Brief History of Data Visualization. Friendly. 2008.  
<http://www.datavis.ca/milestones>